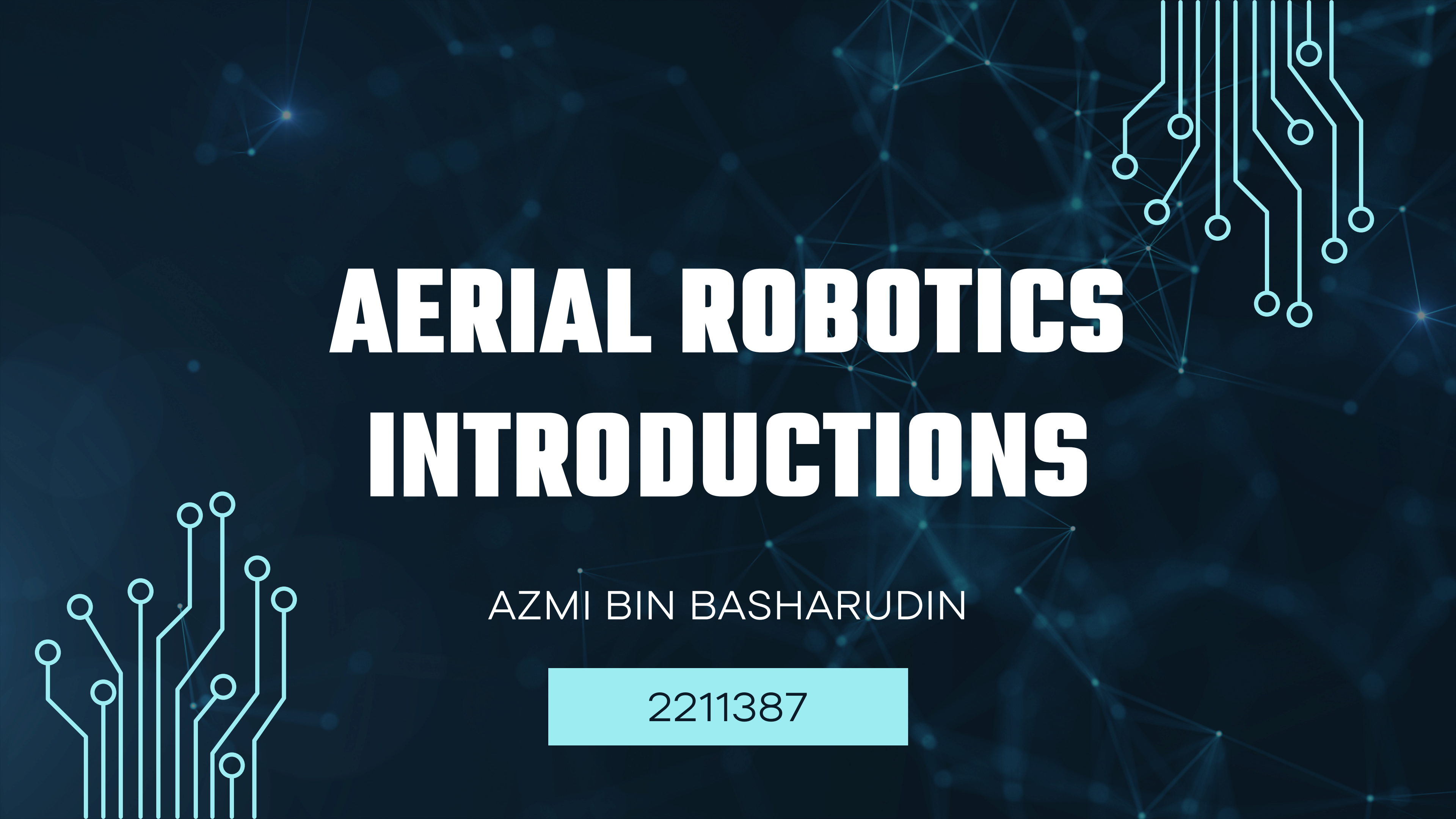




# **AERIAL ROBOTICS INTRODUCTIONS**

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# History



## Early Concepts (Pre-20th Century):

- Ideas of autonomous machines date back to Leonardo da Vinci's mechanical knight.

## Military Origins:

- WWII saw the introduction of rudimentary remote-controlled systems (e.g., radio-controlled aircraft, mines).





# History



## Cold War and Space Era:

- The rise of satellites, missile guidance systems, and early UAVs (Unmanned Aerial Vehicles).

## Late 20th Century:

- Ground and underwater unmanned systems developed for reconnaissance and bomb disposal.

## 21st Century:

- Robotics used in agriculture, delivery, medical, industrial automation.
  - Widespread use of drones, self-driving cars, and smart robots.
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# Current Trends



- AI-enhanced navigation & obstacle avoidance.
- Autonomous mission execution.
- Use in smart cities for delivery, law enforcement, and environmental monitoring.
- Swarming and cooperative drones.
- Integration with 5G and IoT.
- eVTOL (electric vertical take-off and landing) for urban air mobility.





# Manned vs. Unmanned



## Manned:

Requires human pilot/operator.

Better judgment in complex environments.

Risk to human life in hazardous missions.

## Unmanned:

No onboard humans; controlled remotely or autonomously.

Ideal for repetitive, dangerous, or remote tasks.

Lower operating cost, scalable.





**Thank You!**

